PRINTABLE

SIP call-flow cheat sheet

Registration

1. REGISTER to the domain (Request-URI), To = AOR, Contact = device.
2. 401 and WWW-Authenticate. A proxy would send 407 instead.
3. REGISTER with Authorization. New CSeq.
4. 200 OK stores the binding. Expires: 0 removes one. Contact: * and Expires: 0 removes all.

Basic call

1. INVITE (early offer puts SDP here).
2. 100 Trying is hop-by-hop.
3. 180 Ringing is alerting. A To tag means an early dialog, not answer.
4. 200 OK confirms the dialog. ACK for that 2xx is a new end-to-end transaction.
5. BYE ends it. Record-Route keeps the proxy on ACK and BYE.

ACK for a non-2xx final response stays inside the INVITE transaction.

Hold (RFC 3264)

re-INVITE, same Call-ID, higher CSeq. sendonly is answered recvonly. inactive is answered inactive. Port 0 rejects the stream; it is not hold. Avoid c=0.0.0.0.

Transfer

REFER, then 202. 202 is not success. NOTIFY carries message/sipfrag status. Blind: Refer-To is a URI. Attended: Refer-To includes Replaces (Call-ID and tags). RFC 3515, RFC 3891, RFC 5589.

Early media and PRACK

180 with no SDP: local ringback. 183 with SDP: play the RTP. PRACK acknowledges a reliable 1xx. RAck echoes RSeq plus the INVITE CSeq. The 200 to PRACK is not the 200 to INVITE.

Refresh

re-INVITE or UPDATE. Session-Expires and refresher come from RFC 4028. A rejected re-INVITE leaves the last good session in place. 491 means glare.

UC Lab field card · fictional examples only · confirm RFCs and current vendor docs before production changes.

Sources and go deeper

Sources and go deeper
Opened 2026-10. A row marked confirm has no link, because that page was not verified.
RFC. RFC 3261, SIP. https://www.rfc-editor.org/rfc/rfc3261. last checked 2026-10
RFC. RFC 3550, RTP. https://www.rfc-editor.org/rfc/rfc3550. last checked 2026-10
RFC. RFC 4566, SDP. https://www.rfc-editor.org/rfc/rfc4566. last checked 2026-10
Cisco. Cisco Unified Communications Manager product support.
https://www.cisco.com/c/en/us/support/unified-communications/unified-communications-manager-callmanager/series.html. last checked 2026-10
Cisco. System Configuration Guide for CUCM, Release 15.
https://www.cisco.com/c/en/us/td/docs/voice_ip_comm/cucm/admin/15/systemConfig/cucm_b_system-configuration-guide-15.html. last checked 2026-10
Protocol and ports. CUCM 15 TCP and UDP port usage.
https://www.cisco.com/c/en/us/td/docs/voice_ip_comm/cucm/admin/15/systemConfig/cucm_b_system-configuration-guide-15/cucm_m_tcp-and-udp-port-usage-12-0.html. last checked 2026-10
Protocol and ports. IANA SIP parameters. https://www.iana.org/assignments/sip-parameters/sip-parameters.xhtml. last checked 2026-10
Protocol and ports. IANA RTP parameters. https://www.iana.org/assignments/rtp-parameters/rtp-parameters.xhtml. last checked 2026-10
Community. Cisco Community, Collaboration knowledge base.
https://community.cisco.com/t5/collaboration-knowledge-base/tkb-p/4691-docs-collaboration-voice-video. last checked 2026-10
Wikipedia. Session Initiation Protocol (Wikipedia).
https://en.wikipedia.org/wiki/Session_Initiation_Protocol. last checked 2026-10
Wikipedia. Real-time Transport Protocol (Wikipedia).
https://en.wikipedia.org/wiki/Real-time_Transport_Protocol. last checked 2026-10
Cisco. Collaboration 12 SRND book. confirm. The historical Collaboration 12 SRND path opened as a product listing, not the design book. Use the Preferred Architecture index, which was opened. last checked 2026-10
RFC. RFC 3264, offer/answer. https://www.rfc-editor.org/rfc/rfc3264. last checked 2026-10
Wikipedia. Session Description Protocol (Wikipedia).
https://en.wikipedia.org/wiki/Session_Description_Protocol. last checked 2026-10
Wikipedia. Skinny Client Control Protocol (Wikipedia).
https://en.wikipedia.org/wiki/Skinny_Call_Control_Protocol. last checked 2026-10
Wikipedia. CUCM (Wikipedia disambiguation). https://en.wikipedia.org/wiki/CUCM. last checked 2026-10
Wikipedia. English Wikipedia article for Cisco Unified Communications Manager. confirm. That title returned HTTP 404. The CUCM disambiguation page is the Wikipedia page that opened. last checked 2026-10
RFC. RFC 3262, reliable provisional responses (PRACK). https://www.rfc-editor.org/rfc/rfc3262. last checked 2026-10